Machine Learning

12-05-26

continued from last session →
step 1-3 are covered

4) compute total error and new weights for datapoints

total error = sum of weights of wrong predictions

for model m1

Total error = TE_1 = weight of d_4 = 0.2

using TE compute 'model say factor' as α

$$\alpha = \frac{1}{2} \log\left(\frac{1-TE}{TE}\right)$$

$$TE \propto \frac{1}{\alpha} \propto \frac{1}{\text{model performance}} \propto \frac{1}{\text{model importance}}$$

model with higher TE will be given less importance

for m1

$$\alpha_1 = \frac{1}{2} \log\left(\frac{1-TE_1}{TE_1}\right)$$

$$= \frac{1}{2} \log\left(\frac{1-0.2}{0.2}\right)$$

$$\alpha_1 = 0.3$$

Update weights of datapoints using α_1

for wrong predictions

$$w_2' = w_1 \times e^{\alpha_1}$$

for correct predictions

$$w_2' = w_1 \times e^{-\alpha_1}$$

Apply weight normalization after this

$$w_2 = \frac{w_2'}{\sum w_2'}$$

	x_1	x_2	x_3	y	$\hat{y}_1$	w_1	w_2'		$w_2 = w_2'/\sum w_2'$
d_1				C_1	C_1	0.2	$0.2 \times e^{-0.3}$	= 0.15	0.15/0.87 = 0.17
d_2				C_1	C_1	0.2	$0.2 \times e^{-0.3}$	= 0.15	0.15/0.87 = 0.17
d_3				C_2	C_2	0.2	$0.2 \times e^{-0.3}$	= 0.15	0.15/0.87 = 0.17
d_4				C_3	C_2	0.2	$0.2 \times e^{0.3}$	= 0.27	0.27/0.87 = 0.31
d_5				C_2	C_2	0.2	$0.2 \times e^{-0.3}$	= 0.15	0.15/0.87 = 0.17
					$\sum w_1 = 1$		$\sum w_2' = 0.87$		$\sum w_2 = 1$

Data with new weights

	x_1	x_2	x_3	y	w_2
d_1				C_1	0.17
d_2				C_1	0.17
d_3				C_2	0.17
d_4				C_3	0.31
d_5				C_2	0.17

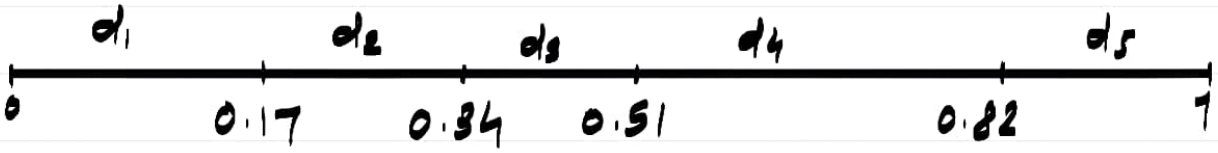
→ the datapoints where previous model made mistakes will get higher weight

5) Create a subset of dataset using weighted sampling

- select 'n' datapoints with weighted sampling

n : total no. of datapoints : 5 (in our case)

	w_2	cummulative w_2
d_1	0.17	0.17
d_2	0.17	0.17 + 0.17 = 0.34
d_3	0.17	0.34 + 0.17 = 0.51
d_4	0.31	0.51 + 0.31 = 0.82
d_5	0.17	0.82 + 0.17 = 0.99 ≈ 1



the datapoint having higher weight gets larger region in weighted sampling

Weighted sampling

1. generate a random number betⁿ 0 to 1
   - using random number generator

   lets say we generate $v_1 = 0.42$

2. pick datapoint for new dataset

   $v_1 : 0.42 \rightarrow$ in region of d_3 – pick d_3 in new dataset

3. repeat step 1 and 2 'n' number of times

   n=5 here

$V_1 = 0.42 \rightarrow$ pick d_3

$V_2 = 0.54 \rightarrow$ pick d_4

$V_3 = 0.65 \rightarrow$ pick d_4

$V_4 = 0.2 \rightarrow$ pick d_2

$V_5 = 0.76 \rightarrow$ pick d_4

Due to weighted sampling, datapoints having higher weights will be picked more number of times in new dataset

$X_1 = [d_1, d_2, d_3, d_4, d_5]$

after weighted sampling using w_2

new dataset $X_2 = [d_3, d_4, d_4, d_2, d_4]$

6) Train next model on new dataset

next model m_2 will be trained on X_2

- As X_2 has multiple duplicates of d_4, the training of m_2 will get biased towards d_4
- hence m_2 will focus more on d_4 → which is mistake from m_1

$X_2 = [d_3, d_4, d_4, d_2, d_4]$

	x_1	x_2	x_3	y		$\hat{y}_2$
d_3				c_2	train model m_2 DTree clf	c_1 ✗
d_4				c_3		c_3 ✓
d_4				c_3		c_3 ✓
d_2				c_1		c_2 ✗
d_4				c_3		c_3 ✓

7) Assign equal weights to datapoints in new dataset

	x_1	x_2	x_3	y	$\hat{y}_2$		w_2
d_3				C_2	C_1	✗	0.2
d_4				C_3	C_3	✓	0.2
d_4				C_3	C_3	✓	0.2
d_2				C_1	C_2	✗	0.2
d_4				C_3	C_3	✓	0.2

} scrap previous w_2 values and assign equal weights

8) Repeat steps 4 to 7 untill the specified number of base learner models are trained

	x_1	x_2	x_3	y	$\hat{y}_2$		w_2
d_3				C_2	C_1	✗	0.2
d_4				C_3	C_3	✓	0.2
d_4				C_3	C_3	✓	0.2
d_2				C_1	C_2	✗	0.2
d_4				C_3	C_3	✓	0.2

step 4 →

$$TE_2 = 0.2 + 0.2 = 0.4$$

$$\alpha_2 = \frac{1}{2}\left(\frac{1 - TE_2}{TE_2}\right) = \frac{1}{2}\left(\frac{0.6}{0.4}\right) = 0.09$$

$w_3' = w_2 \times e^{\alpha_2}$ → for wrong predictions

$w_3' = w_2 \times e^{-\alpha_2}$ → for correct predictions

$w_3 = \frac{w_3'}{\sum w_3'}$ → normalize the weights

Step 5 → weighted sampling

$X_3 \rightarrow [\, d_3 \;\; d_2 \;\; d_3 \;\; d_2 \;\; d_4 \,]$

Step 6 → train model m_3 on dataset X_3

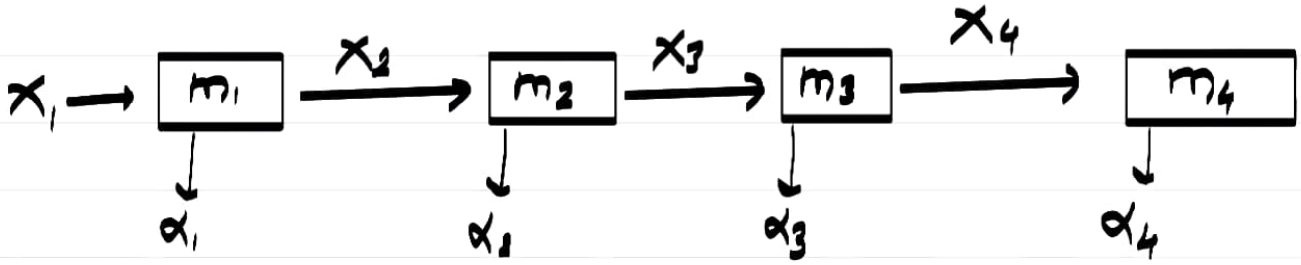
	x_1	x_2	x_3	y	$\hat{y}_3$		w_3	
d_3				c_2	c_2	✓	0.2	} Step 7
d_2				c_1	c_1	✓	0.2	
d_3				c_2	c_2	✓	0.2	
d_2				c_1	c_1	✓	0.2	
d_4				c_3	c_2	✗	0.2	

We keep on repeating step 4 → step 7+8 until

1. required no. of base learner models are trained
2. if errors of the subsequent models are not changing much

Model predictions after training:

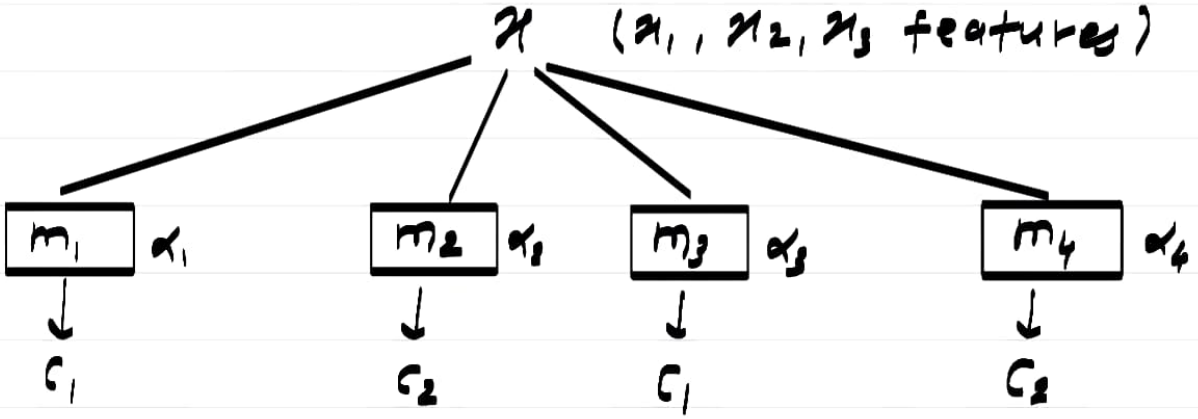
lets say we trained 4 DTrees in Adaboost classifier



new datapoint

$x \rightarrow$

new datapoint



let's say $\alpha_1 = 0.6$

$\alpha_2 = 0.75$

$\alpha_3 = 0.4$

$\alpha_4 = 0.2$

for final prediction

$\hat{y}$	supporting models	combined α
c_1	m_1, m_3	$\alpha_1 + \alpha_3 = 0.6 + 0.4 = 1$
c_2	m_2, m_4	$\alpha_2 + \alpha_4 = 0.75 + 0.2 = 0.95$
c_3	–	0

combined α for c_1 > combined α for c_2

final prediction = c_1

